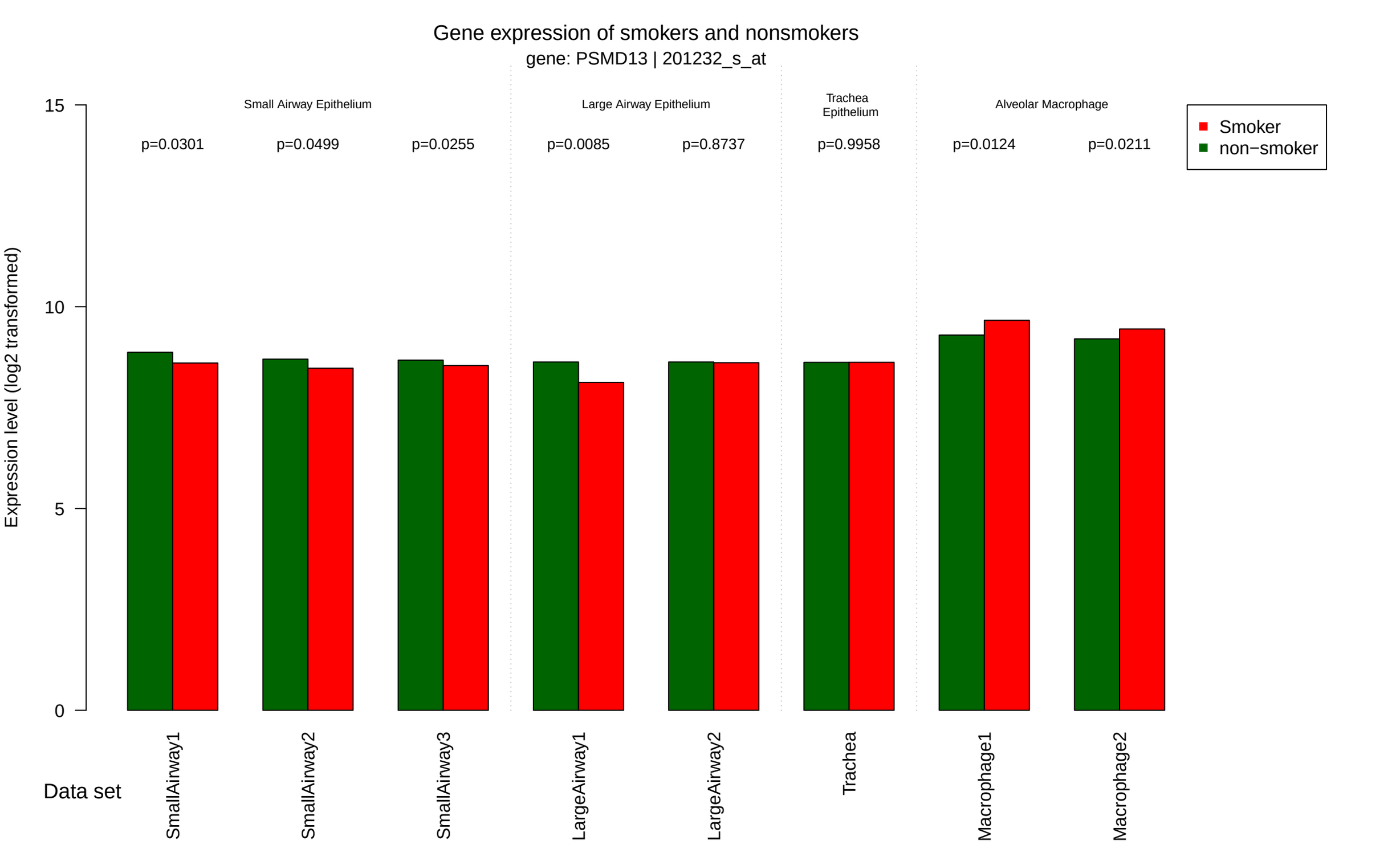
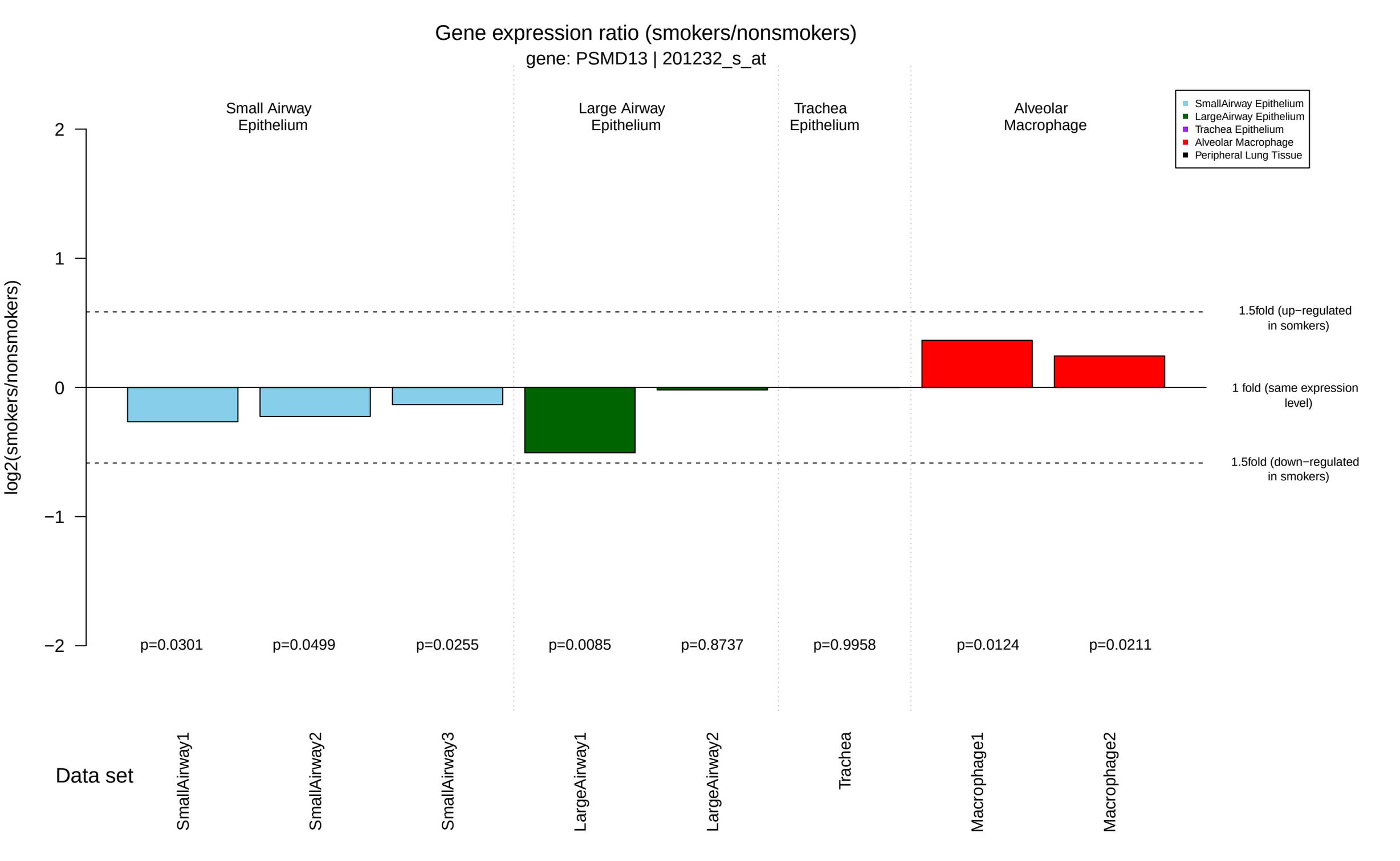
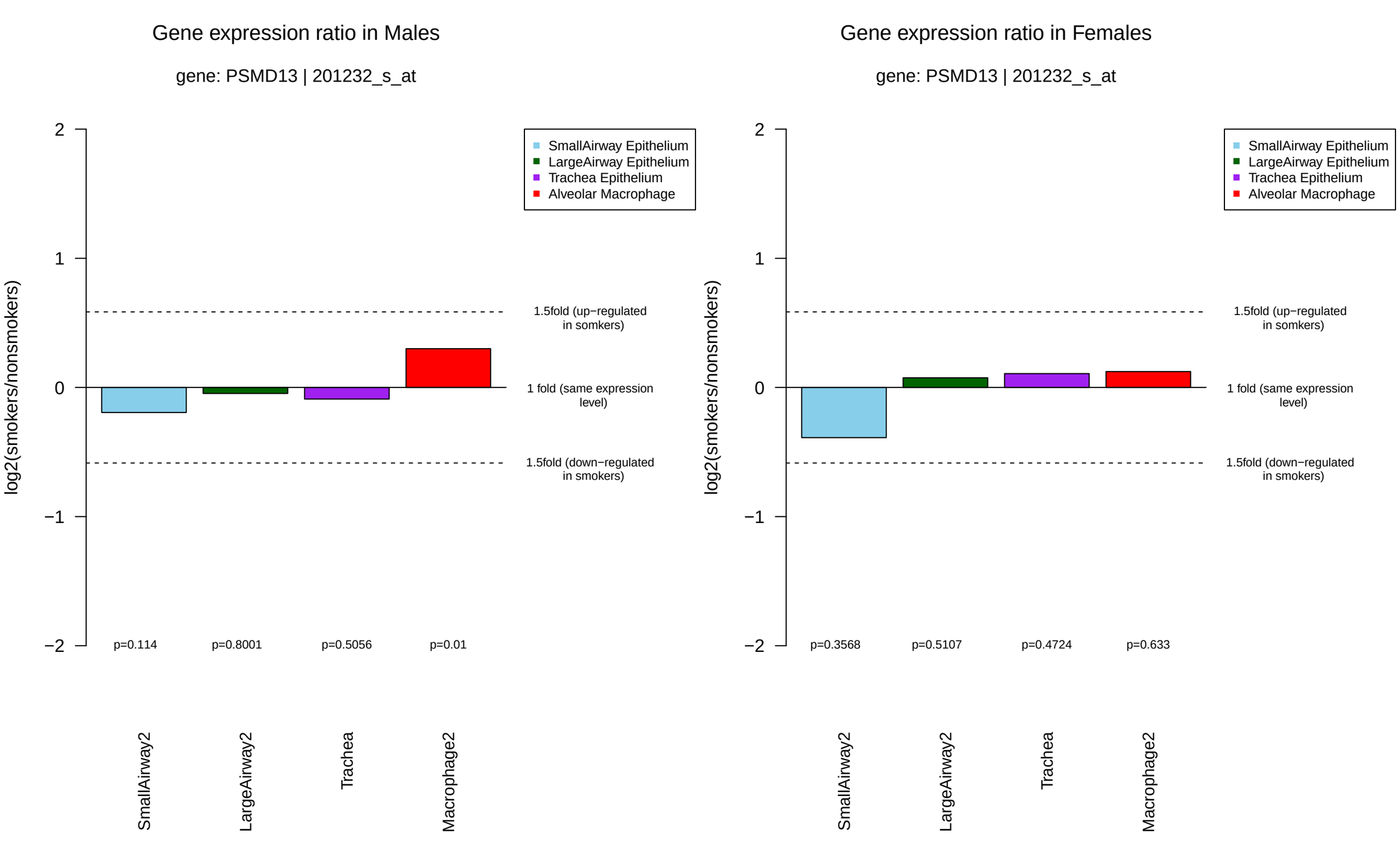
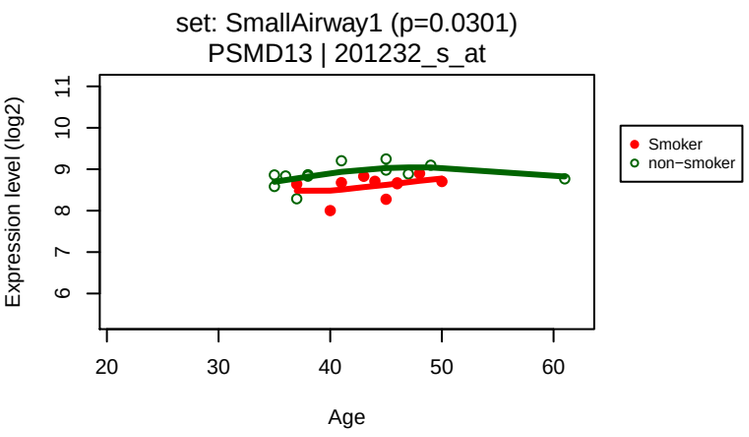
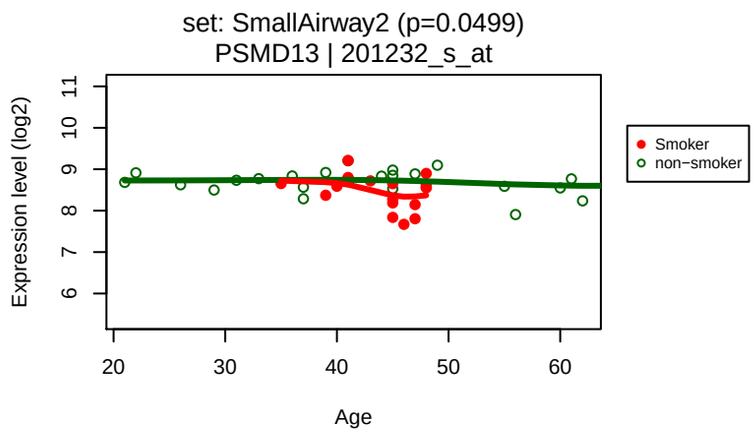
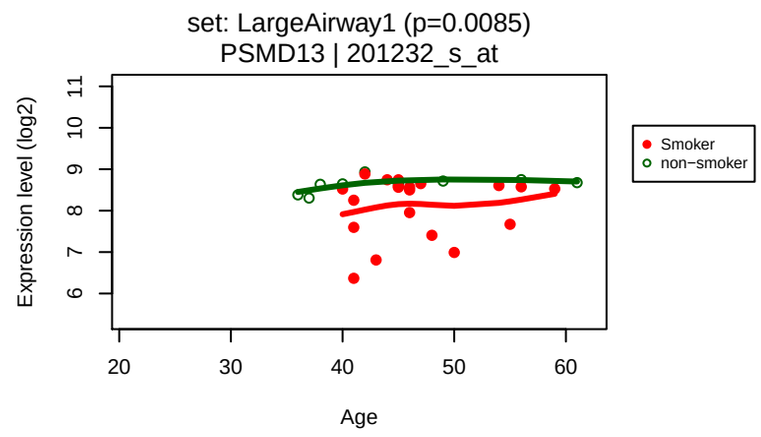

Gene expression of smokers and nonsmokers
gene: PSMD13 | 201232_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0301
p=0.0499
p=0.0255
p=0.0085
p=0.8737
p=0.9958
p=0.0124
p=0.0211
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

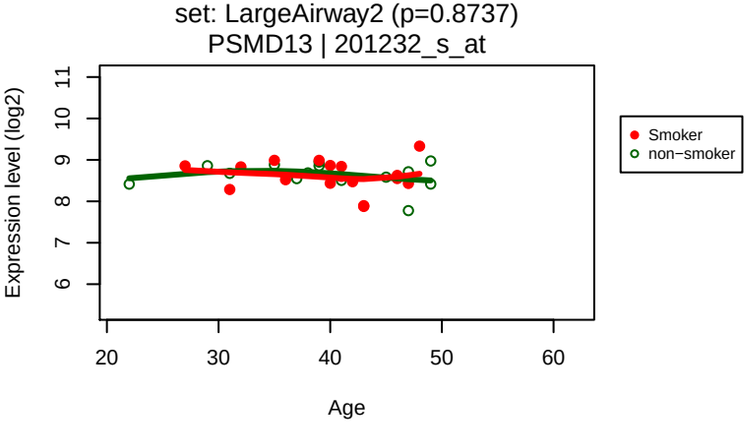

Gene expression ratio (smokers/nonsmokers)
gene: PSMD13 | 201232_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0301
p=0.0499
p=0.0255
p=0.0085
p=0.8737
p=0.9958
p=0.0124
p=0.0211
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

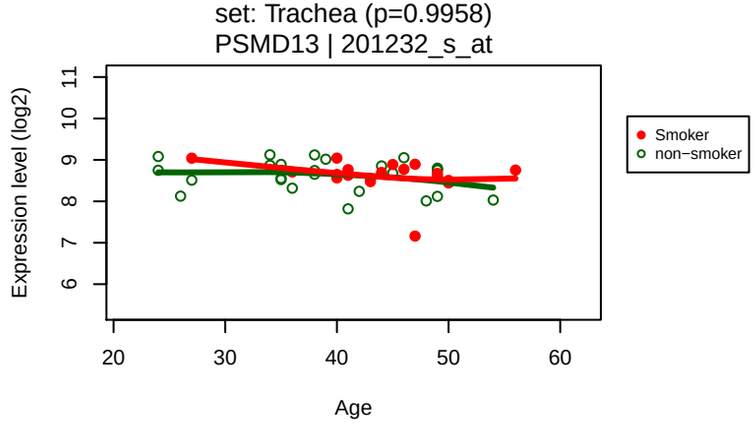

Gene expression ratio in Males
gene: PSMD13 | 201232_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.114
p=0.8001
p=0.5056
p=0.01
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PSMD13 | 201232_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3568
p=0.5107
p=0.4724
p=0.633
SmallAirway2
LargeAirway2
Trachea
Macrophage2

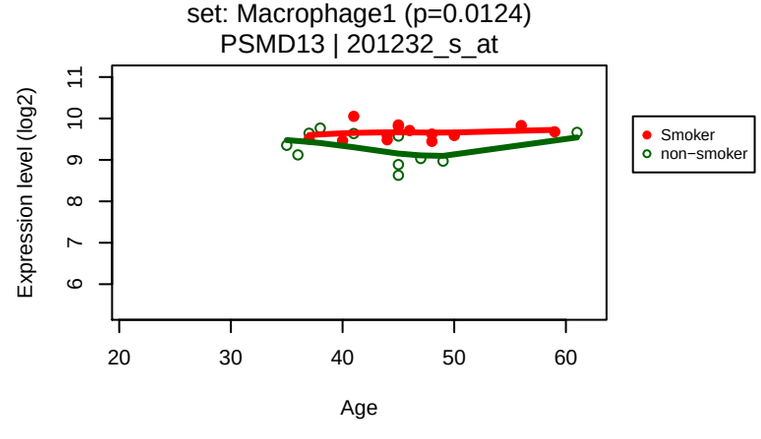

set: SmallAirway1 (p=0.0301)
PSMD13 | 201232_s_at
Expression level (log2)
Age
Smoker
non−smoker

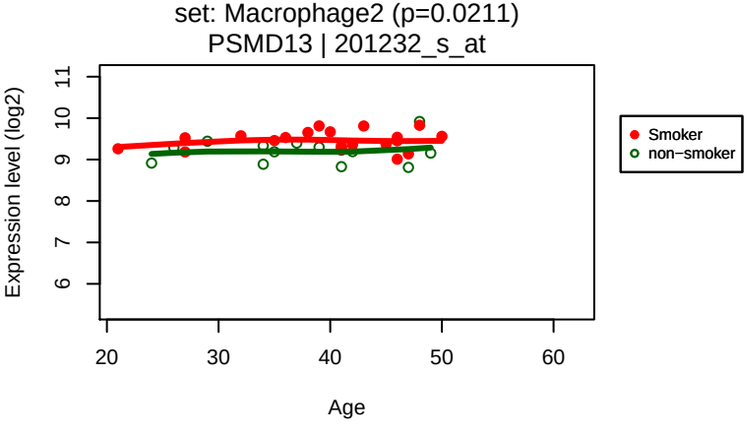

set: SmallAirway2 (p=0.0499)
PSMD13 | 201232_s_at
Expression level (log2)
Age
Smoker
non−smoker

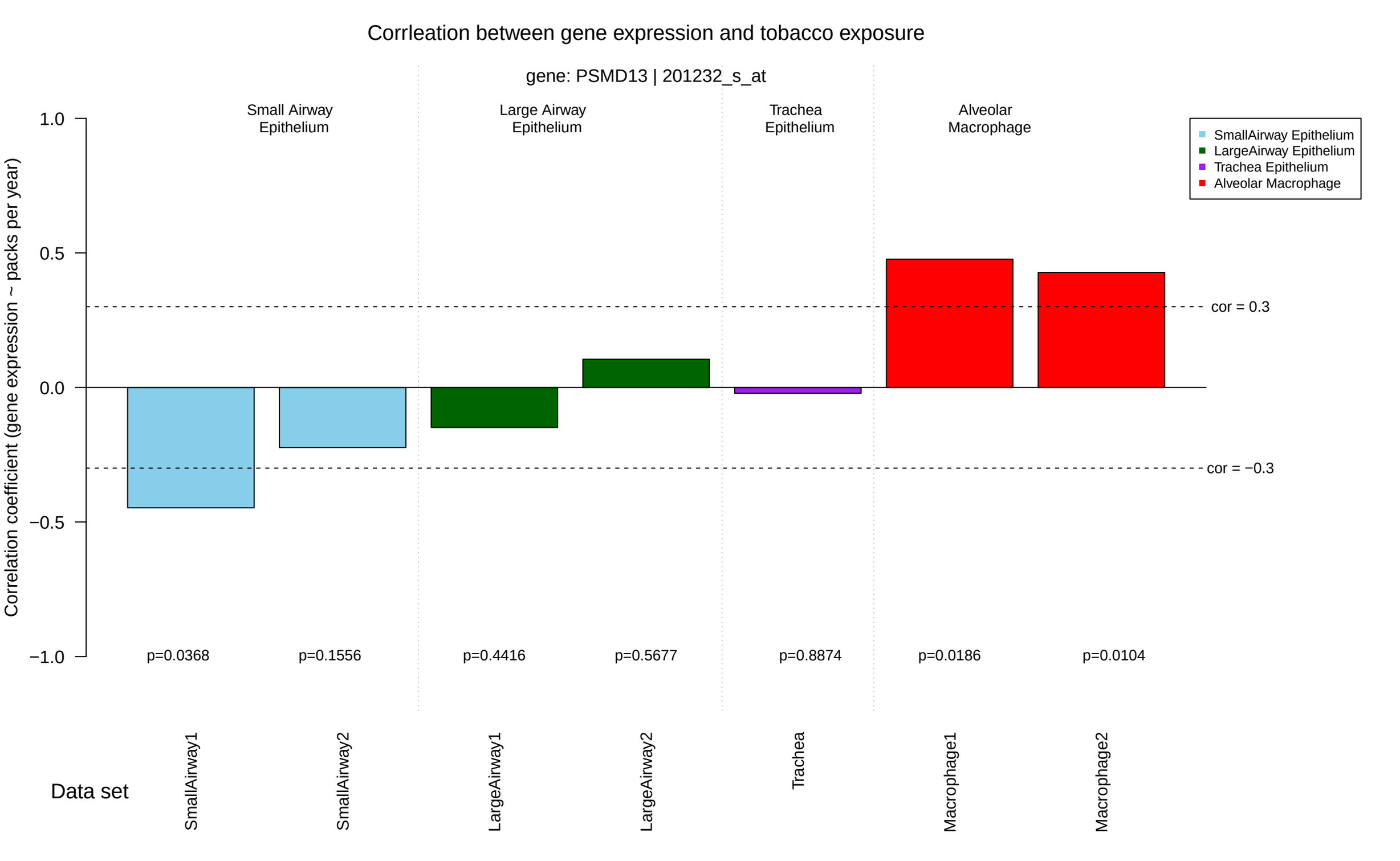

set: LargeAirway1 (p=0.0085)
PSMD13 | 201232_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8737)
PSMD13 | 201232_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9958)
PSMD13 | 201232_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0124)
PSMD13 | 201232_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0211)
PSMD13 | 201232_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PSMD13 | 201232_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0368
p=0.1556
p=0.4416
p=0.5677
p=0.8874
p=0.0186
p=0.0104
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2